

Final Project Potato Slicer

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EML 2023

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November 28, 2023

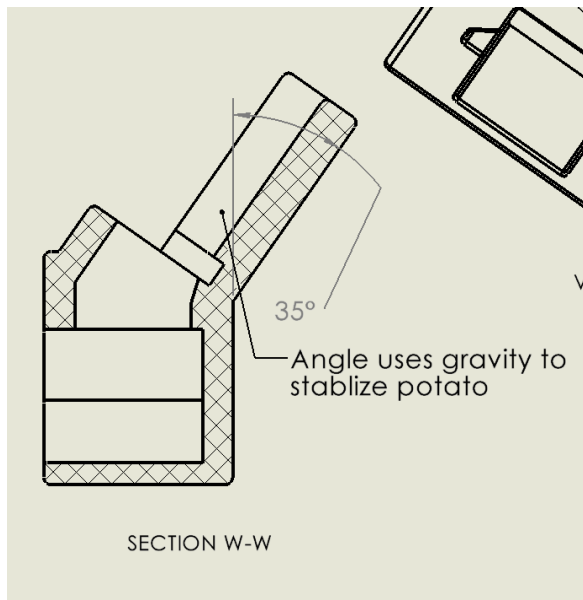
Objective

The overall objective of my slicer was to cut potatoes into 1/4-inch French fries in a fast and efficient manner. My design aimed to cut potatoes of small to medium size, ranging from 1.5 to 3 inches (38 to 76 millimeters) in diameter. My slicer must fit in a home kitchen environment, being small enough to be stored under a cabinet like a mixer or food processor. Another one of my design parameters is that the potatoes must be directly deposited into a storage container after being sliced. This storage container, along with the cutting blades, must be detachable from the main body. Also, the storage container, blades, and any other removable component must be both durable and dishwasher-safe. Furthermore, the vegetable slicer will require little assembly and contain very few parts. Less parts and assembly means fewer places for dirt and grime to collect, resulting in less overall cleaning of the slicer. In addition, the “cutting” action of my slicer must be user-powered; my design should refrain from using levers, gears, or other complex mechanisms. Overall, I aimed to design an intuitive, multi-component, hand-powered potato slicer capable of cutting French fries that required very little assembly with a relatively simple but attractive design.

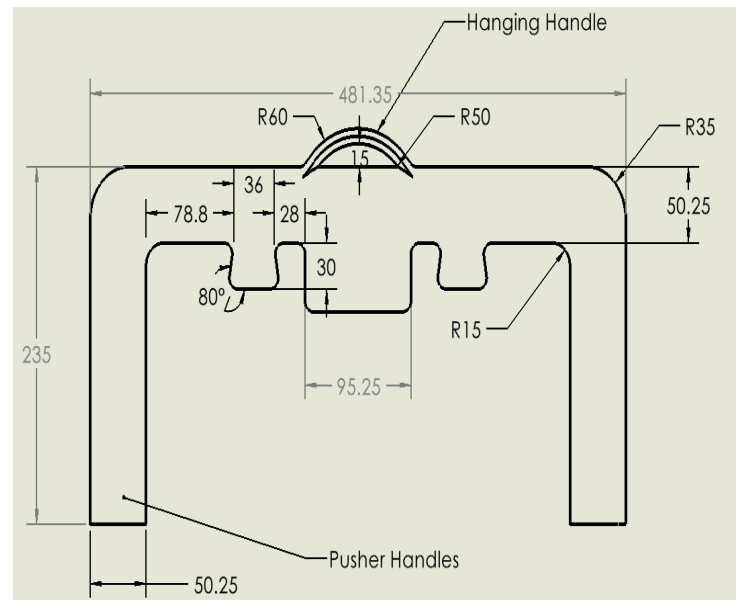
Concept Selection

The guillotine served as my initial design inspiration because I imagined the blades slamming down on the potatoes to cut them. I felt this was good because it would be hand-powered and relatively simple to design and assemble with only one moving part. I soon abandoned this idea, as I couldn't think of a way to deposit the slices in a container. I usually try to build my designs in my head before drawing them out. After concluding that I was unable to think of a deposit method, I didn't bother with drawing a physical design.

I then revised my initial idea into a reverse guillotine design, in which a block would push the vegetable down upon the blades, passively cutting them into fries. In contrast to my original design, which involved a blade actively cutting the potatoes. This plate would sit in a slot to help guide its downward motion. My initial design uses square slots as I thought they were simple; however, I later switched to a dovetail slot design as I felt it would guide the plate better. This plate also has a handle, which allows it to be stored on the wall for easy storage. Furthermore, I decided to have my cutting mechanism at a 35-degree angle to the top plane. This option provided a place for the potatoes to rest while being sliced. If the cutting mechanism was completely vertical, the potatoes could fall out or be misaligned. With this angle, the slicer could rely on gravity to align the vegetables and keep them in place.



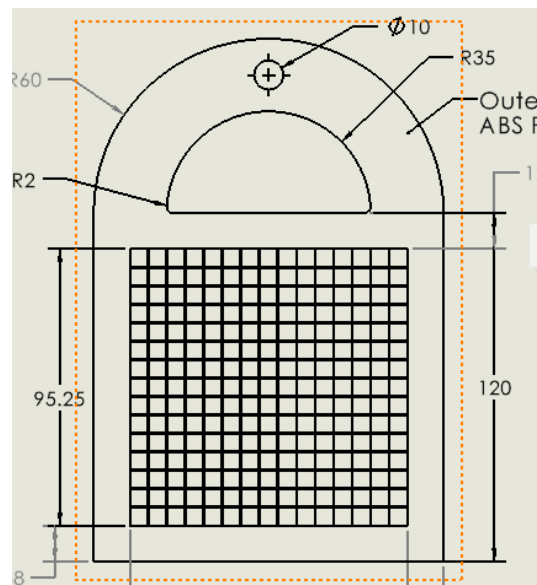
Main Body Component CAD Drawing



Pusher Plate CAD Drawing

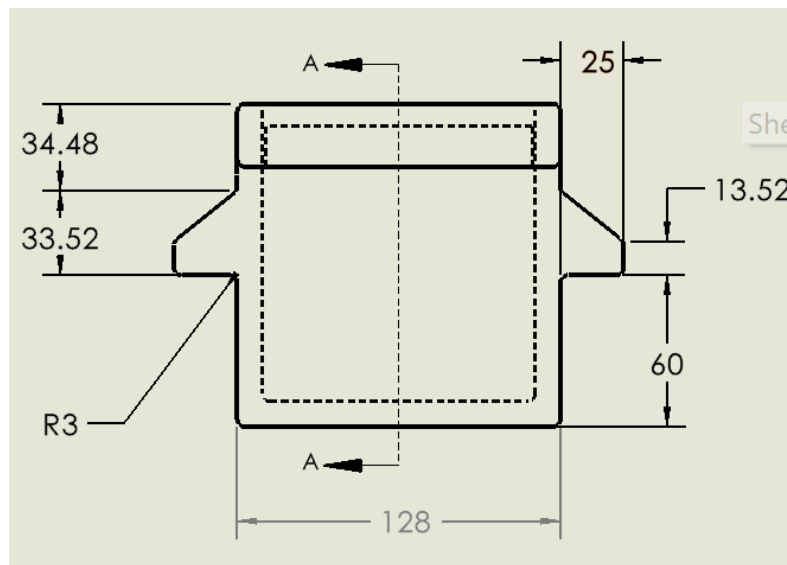
Moreover, I later added springs to my design, located in slots that the pusher plate runs on. I unfortunately didn't include this in my initial concept. The springs' purpose is to return the pusher plate to the top after slicing the potatoes, meaning the user doesn't have to manually pull the plate back. Pushing is naturally easier than pulling for humans so I felt it was the right choice to have the spring perform the more laborious job.

Moreover, I aimed for my blades to be both removable and dishwasher-safe. To accomplish this goal, I went with a rectangular slot design that would have the blades line up directly with the back face of the slicer. The blades are fashioned in a square grid pattern, with each square being approximately $\frac{1}{4}$ -inch in length. The blade component also sits at an angle with the pusher mechanism, relying on both gravity and friction to keep it aligned and in place. I settled on making the blade component have a square shape with a handle at one end so it could be easily inserted into its slot. The handle also has a hole so that the blade could be stored on a wall. This made the blade component not only convenient to clean but also to store.



Blade Component

Moving on, the basket components shared many of the same goals as the blade component, particularly needing to be both removable and dishwasher-safe. I also focused on the basket's size and strength, as I wanted it to be large and strong enough to hold a sufficient amount of fries. After a couple of sketches, I decided on using a drawer design for my basket. The basket would sit in a slot and could be inserted and removed from the main body with ease. It would also have a handle with a semi-circle undercut to aid in removing the basket from the main body. This basket would be stored along with the main body. I felt this design was both a simple and effective method for depositing the cut potatoes in a storage container.

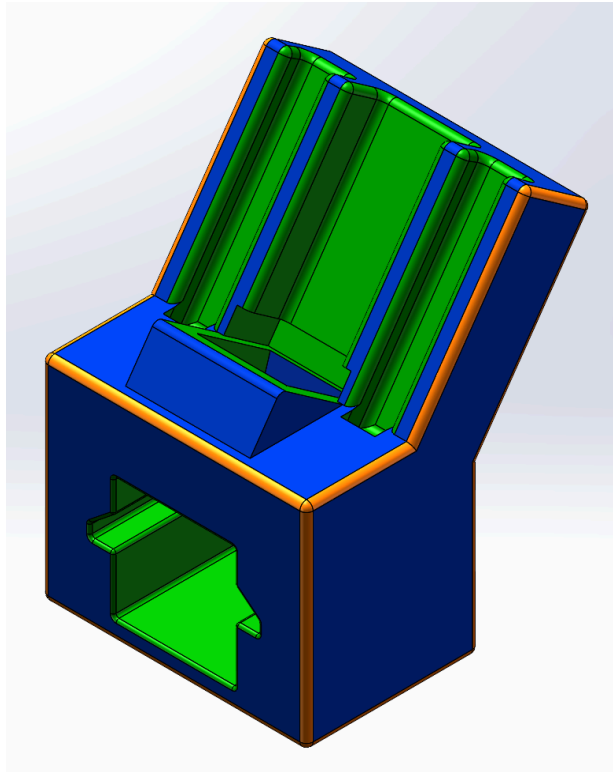


Basket Component

Functionality (Assembly)

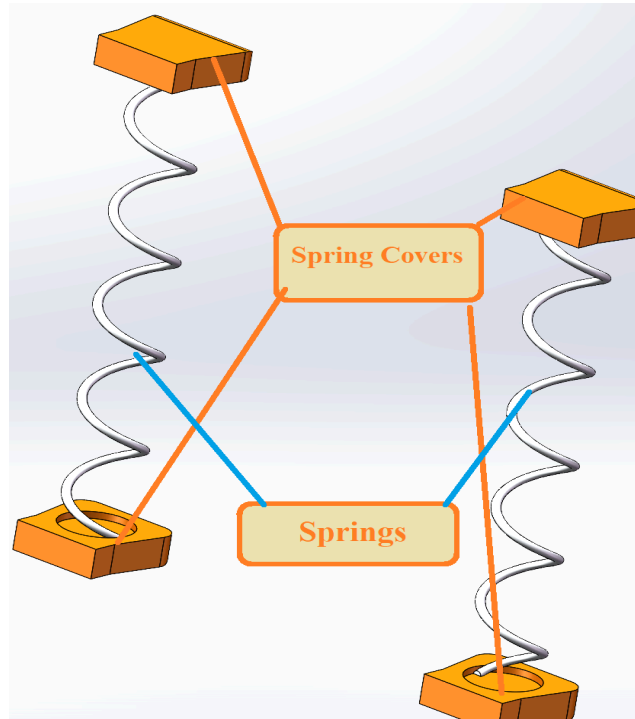
As mentioned before, one of my overall design goals was simple and easy assembly. The first and largest part of the assembly is the Main Body component, which acts as the base for the

entire vegetable slicer. Because of its shape, I imagine that it would be manufactured via injection molding or 3D printing with ABS as the material.

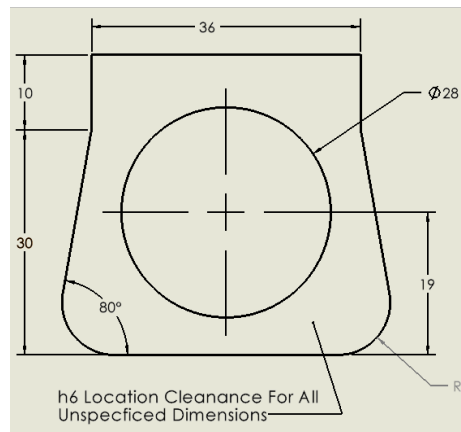


Main Body Component

Next are the springs and the spring covers, the only components that require assembly. The springs are designed to be bought, while their covers would be manufactured with injection molding. By placing a spring cover on the end of each spring, the return spring sub-assembly is done. As mentioned before, this sub-assembly returns the pusher plate back to the top after each cut.



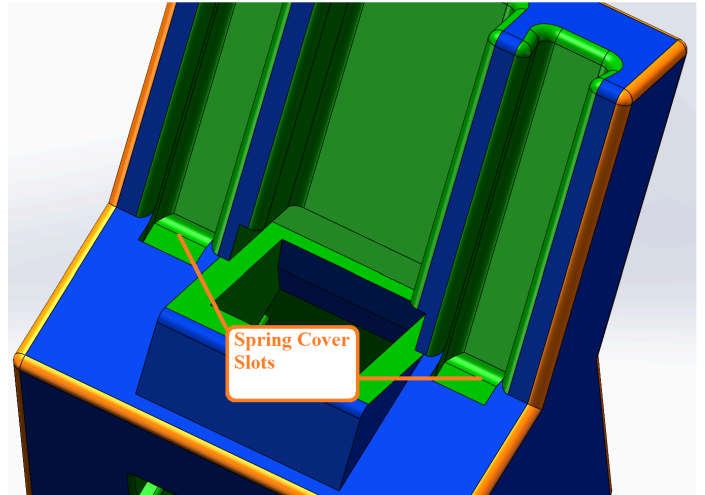
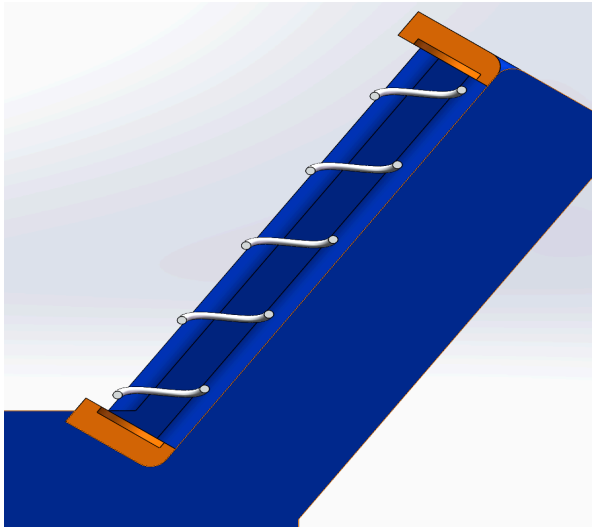
Return Assembly



Spring Cover CAD Drawing

To finish this assembly, each of the springs will be placed in its associated slot located on opposite sides of the Main Body component. To ensure they don't become loose from wear and detach from the main body, a strong adhesive such as super glue will be applied at the bottom of

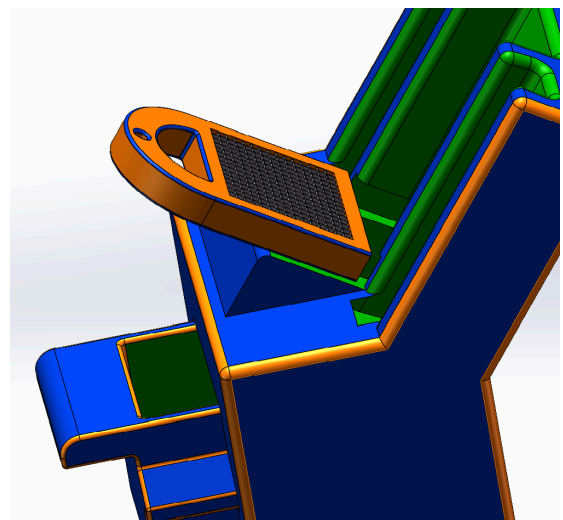
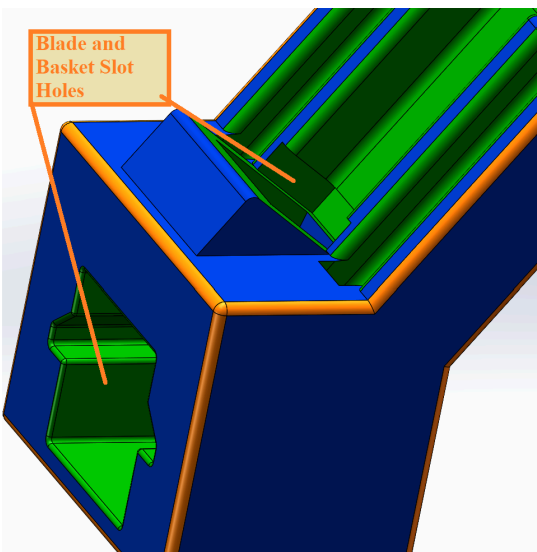
each lower spring cover. They are designed to be press-fit and may require some force from a mallet to install.



Springs Inside Slot

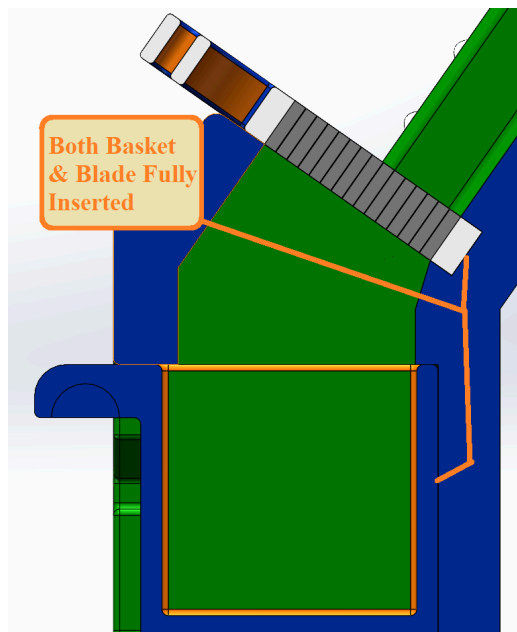
With the return spring system installed, the last steps of the assembly can be completed. The assembly involves both the basket and blade components. Both should be placed in their dedicated slots with ease. There is free-running clearance between the components and their associated holes so they should slide in with ease. And as mentioned before, both are designed to be removable from the main body and slide out with ease as well.

Spring Cover Slots

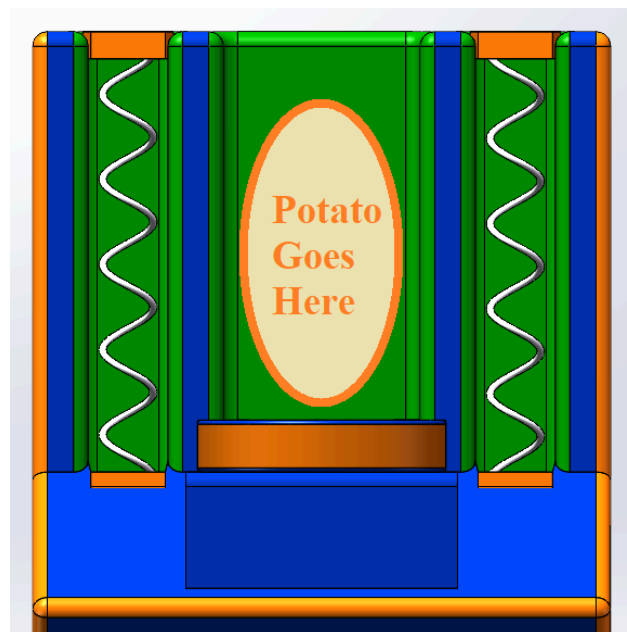


Functionality (Operation)

As previously stated, this slicer employs a reverse guillotine design, in which a force pushes the potatoes down onto the blades, passively cutting them into fries. To operate the slicer, first ensure that both the basket and blade components are fully inserted into their slots, with their back ends in contact with the main body component's wall. Next, place a potato of small to medium size in the square slot located in the middle of the main body component; because of the slot's angle and gravity, the potato should sit safely inside its slot.



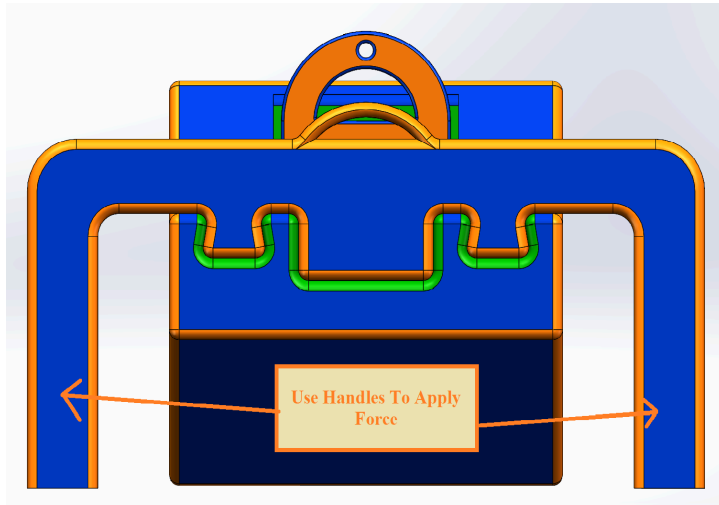
Fully Inserted Basket and Blade



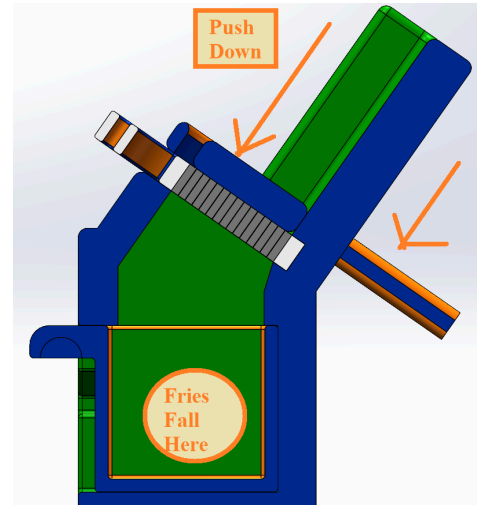
Auxiliary View of Potato Slot

Next, align the pusher plate so that it aligns with the slots and spring covers located at the top of the main body. Using the handles, constant, downward force must be applied using the pusher plate until the potato is fully cut. As the potato is being cut, the springs should help guide

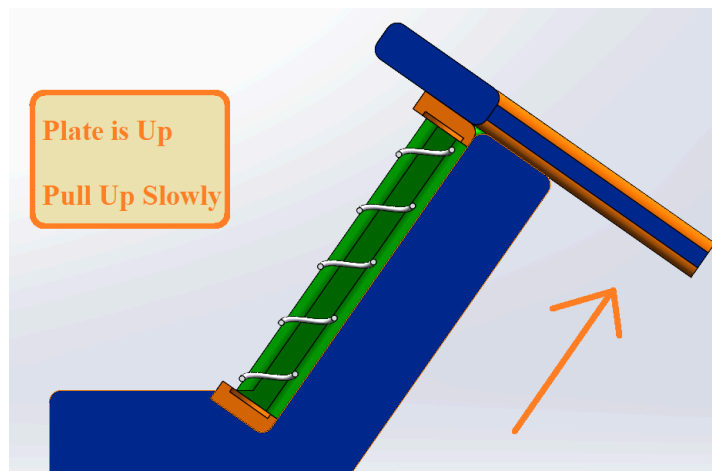
the plate down its path, with the springs being compressed along the way. After the potato is fully cut, its slices should fall into the basket below. Then, the plate should be slowly lifted back up its path, allowing the spring to guide its way. This process is then repeated until the basket is full and its contents are emptied.



Aligning Pusher Plate



Fries Being Cut



Pulling Plate Back Up

Conclusion

Through the designing and modeling process, I made a series of decisions relating to component material, component manufacturing, material cost, and overall assembly and usage. Each of these choices had its advantages and disadvantages that affected the overall project. For example, the main body component would have to be 3D-printed and not injection-molded because of its complex geometry; this will in turn hinder its ability to be mass produced. However, its geometry gives it a unique look and allows it to take advantage of gravity to align components. Plus, its removable component design allows for easy cleaning and replacement. Furthermore, its dovetail slots and spring return system help with the alignment and stability of the pusher plate, making the slicing action and overall experience smoother.

Moving on, all of the components (except the springs) are composed of ABS (acrylonitrile butadiene styrene) plastic with a PC (polycarbonate) coating. ABS was chosen because it is a fairly common material used in food manufacturing products, meaning its cost is relatively low. ABS is also a strong thermoplastic that can be used in both injection molding and 3D-printing manufacturing. However, on its own, ABS is occasionally toxic (depending on the variety); a PC coating was chosen to remove this disadvantage, making the slicer food safe. Furthermore, both the springs and blades of the components are composed from 304 Stainless steel because of its low cost, heat and corrosive resistance, and wide use in the food production industry.

Moreover, there are some other cons to my slicer design that may cause problems. First, the blades of my slicer are passive cutters, meaning they don't play an active role in cutting the material. Potatoes are hardy vegetables, meaning sufficient physical strength would be needed to operate the slicer. Also, because of the friction, the plastic constructing the various holes and slots of the slicer may wear down and no longer work properly. However, I believe my design is sturdy and will last a significant amount before it succumbs to wear and tear.

In conclusion, I've achieved my overall objective by designing a kitchen-friendly potato cutter that makes French fries in a fast and efficient manner. Along with food-safe, dishwasher-safe removable components made from affordable materials, my slicer utilizes a unique design. I feel my slicer would make a great addition to any home kitchen.